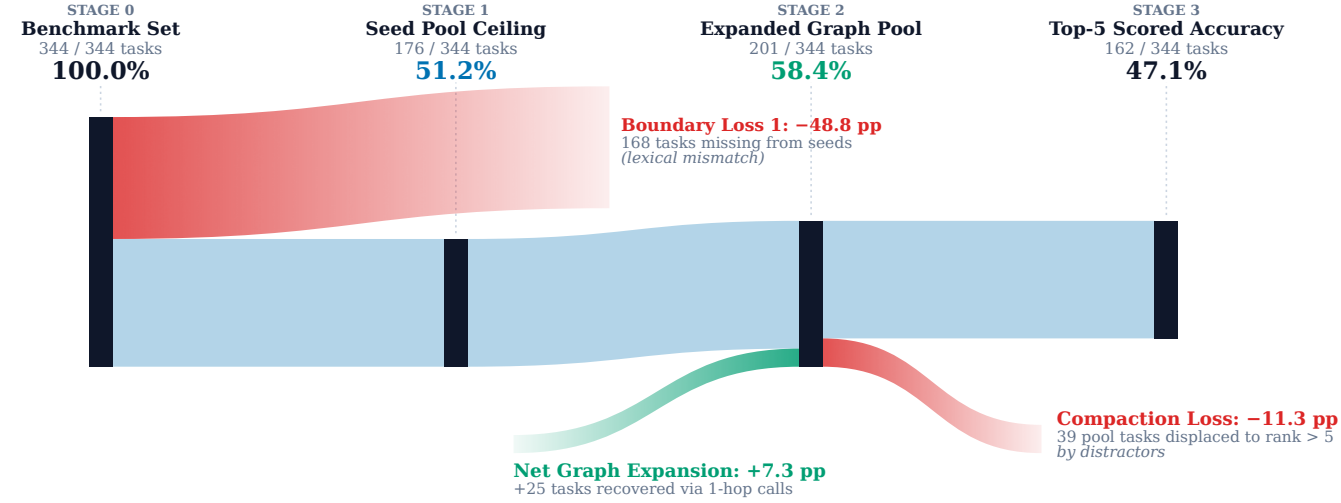


Key Finding: **Upstream Boundary Loss** (41.6 pp net) exceeds **Downstream Compaction Loss** (11.3 pp) by **3.68×** (Bottleneck Ratio). **Retrieval boundary is the primary bottleneck.**